

Geometry & Measurement

ANSWER KEY



Angle facts

- 1 Two angles on a straight line measure $3x$ and $2x + 30$. Find x .
Angles on a straight line sum to 180° : $3x + 2x + 30 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$.
 - 2 Find the angle vertically opposite a 65° angle.
Vertically opposite angles are equal, so the angle is 65° .
 - 3 Three angles around a point measure 110° , 95° , and x . Find x .
Angles around a point sum to 360° : $x = 360 - 110 - 95 = 155^\circ$.
 - 4 Two angles are complementary (sum to 90°). One measures 37° . Find the other.
 $90 - 37 = 53^\circ$.
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Parallel lines and transversals

- 5 Two parallel lines are cut by a transversal, creating an angle of 70° . Find its corresponding angle.
Corresponding angles are equal, so the angle is 70° .
 - 6 In the same setup, find the co-interior (allied) angle to a 70° angle.
Co-interior angles sum to 180° : $180 - 70 = 110^\circ$.
 - 7 Find the alternate interior angle to a 55° angle.
Alternate interior angles are equal, so the angle is 55° .
 - 8 Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x .
Co-interior angles sum to 180° : $4x + 2x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.
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Triangle angle sum

- 9 Find the third angle of the triangle shown, given the two marked angles.
 $180 - 40 - 75 = 65^\circ$.
- 10 An isosceles triangle has two equal angles of 72° each. Find the third angle.
 $180 - 72 - 72 = 36^\circ$.
- 11 In a right triangle, one of the non-right angles is 35° . Find the other non-right angle.
 $180 - 90 - 35 = 55^\circ$.

- 12 A triangle has three equal angles. Find each angle, and name this type of triangle.
 $180 \div 3 = 60^\circ$ each — this is an equilateral (equiangular) triangle.
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Angle sums of polygons

- 13 Find the sum of the interior angles of a hexagon (6 sides).
 $\text{Sum} = (n - 2) \times 180^\circ = (6 - 2) \times 180^\circ = 720^\circ$.
- 14 Find the sum of the interior angles of a pentagon (5 sides).
 $\text{Sum} = (5 - 2) \times 180^\circ = 540^\circ$.
- 15 A regular polygon has an interior angle sum of $1,080^\circ$. How many sides does it have?
 $(n - 2) \times 180 = 1,080 \Rightarrow n - 2 = 6 \Rightarrow n = 8$ sides (a regular octagon).
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The Pythagorean theorem

- 16 Find the hypotenuse of the right triangle shown.
 $c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm.
- 17 A right triangle has hypotenuse 13 and one leg 5. Find the other leg.
 $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.
- 18 A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?
 $\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ m.
- 19 A right triangle has legs 9 and 12. Find the hypotenuse.
 $\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.
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Perimeter and area of rectangles and squares

- 20 A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.
 $\text{Perimeter} = 2(12 + 7) = 38$ cm. $\text{Area} = 12 \times 7 = 84$ cm².
- 21 A square has side length 9 cm. Find its perimeter and area.
 $\text{Perimeter} = 4 \times 9 = 36$ cm. $\text{Area} = 9^2 = 81$ cm².
- 22 A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.
 $\text{Width} = \frac{50}{2} - 16 = 25 - 16 = 9$ cm. $\text{Area} = 16 \times 9 = 144$ cm².

- 23 A square has area 121 cm^2 . Find its side length and perimeter.

$$\text{Side} = \sqrt{121} = 11 \text{ cm. Perimeter} = 4 \times 11 = 44 \text{ cm.}$$

Area of triangles and parallelograms

- 24 A triangle has base 14 cm and height 9 cm. Find its area.

$$\text{Area} = \frac{1}{2} \times 14 \times 9 = 63 \text{ cm}^2.$$

- 25 A parallelogram has base 12 cm and height 8 cm. Find its area.

$$\text{Area} = 12 \times 8 = 96 \text{ cm}^2.$$

- 26 A triangle has area 60 cm^2 and base 15 cm. Find its height.

$$\text{Height} = \frac{2 \times 60}{15} = 8 \text{ cm.}$$

- 27 A parallelogram has area 126 cm^2 and height 9 cm. Find its base.

$$\text{Base} = \frac{126}{9} = 14 \text{ cm.}$$

Circles: circumference and area

- 28 A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.

$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm.}$$

- 29 A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.

$$A = \pi r^2 = \frac{22}{7} \times 49 = 154 \text{ cm}^2.$$

- 30 A circle has diameter 20 cm. Find its circumference in terms of π .

$$C = \pi d = 20\pi \text{ cm.}$$

- 31 Find the area of the shaded 90° sector shown, in terms of π .

$$\text{Sector area} = \frac{90}{360} \times \pi \times 10^2 = \frac{1}{4} \times 100\pi = 25\pi \text{ cm}^2.$$

Composite figures

- 32 Find the area of the L-shaped room shown.

Treat it as a $10 \text{ m} \times 8 \text{ m}$ rectangle with a $4 \text{ m} \times 3 \text{ m}$ notch removed: $10 \times 8 - 4 \times 3 = 80 - 12 = 68 \text{ m}^2$.

- 33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.

$$15 \times 6 - 5 \times 2 = 90 - 10 = 80 \text{ m}^2.$$

- 34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.
Rectangle: $12 \times 5 = 60 \text{ cm}^2$. Triangle: $\frac{1}{2} \times 6 \times 5 = 15 \text{ cm}^2$. Total = $60 + 15 = 75 \text{ cm}^2$.
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Volume and surface area

- 35 A rectangular box has dimensions 5 cm $\times$ 4 cm $\times$ 3 cm. Find its volume.
 $V = 5 \times 4 \times 3 = 60 \text{ cm}^3$.
- 36 A cube has edge length 6 cm. Find its volume and surface area.
 $V = 6^3 = 216 \text{ cm}^3$. Surface area = $6 \times 6^2 = 6 \times 36 = 216 \text{ cm}^2$.
- 37 A rectangular prism has volume 240 cm^3 , length 10 cm, and width 6 cm. Find its height.
Height = $\frac{240}{10 \times 6} = \frac{240}{60} = 4 \text{ cm}$.
- 38 Find the surface area of a rectangular box with dimensions 8 cm $\times$ 5 cm $\times$ 4 cm.
 $SA = 2(lw + lh + wh) = 2(40 + 32 + 20) = 2 \times 92 = 184 \text{ cm}^2$.
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Word problems

- 39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.
Outer dimensions: $22 \text{ m} \times 17 \text{ m} = 374 \text{ m}^2$. Garden area: $20 \times 15 = 300 \text{ m}^2$. Path area = $374 - 300 = 74 \text{ m}^2$.
- 40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.
Area = $6 \times 4 = 24 \text{ m}^2$. Cost = $24 \times 45 = 1,080 \text{ riyals}$.
- 41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?
 $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ m}$.
- 42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.
One lap = $2\pi r = 2 \times \frac{22}{7} \times 35 = 220 \text{ m}$. Four laps = $4 \times 220 = 880 \text{ m}$.